

HKDSE Mathematics Compulsory Part — Paper 2 (Mock)

PAPER 2

75 minutes (1.25 hours)

INSTRUCTIONS

1. Read the instructions on the Answer Sheet carefully. Stick a barcode label after the start of the examination.
2. All questions carry equal marks.
3. ANSWER ALL QUESTIONS. Use an HB pencil to mark all answers on the Answer Sheet.
4. Mark only ONE answer for each question. More than one answer scores NO MARKS.
5. No marks will be deducted for wrong answers.

Not to be taken away before the
end of the examination session

There are 30 questions in Section A and 15 questions in Section B.

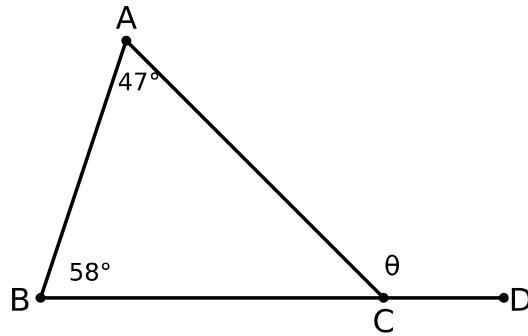
The diagrams in this paper are not necessarily drawn to scale. Choose the best answer for each question.

Section A

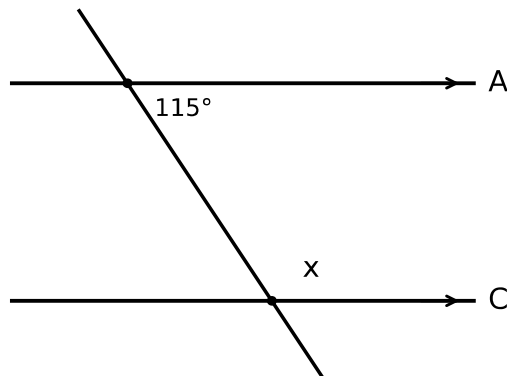
1. Simplify $\frac{a^5b^{-3}}{a^{-1}b^2}$.
A. $\frac{a^6}{b}$
B. $\frac{a^6}{b^5}$
C. $\frac{a^4}{b^5}$
D. a^6b^5
2. Express 0.0000807 in scientific notation.
A. 8.07×10^{-4}
B. 80.7×10^{-6}
C. 8.07×10^5
D. 8.07×10^{-5}
3. Solve $6(x - 3) = 3x + 6$.
A. $x = 6$
B. $x = 24$
C. $x = 4$
D. $x = 8$
4. Simplify $\frac{x^2 - 49}{x^2 - 3x - 28}$.
A. $\frac{x - 7}{x + 4}$
B. $\frac{x + 7}{x - 4}$
C. $\frac{x - 7}{x - 4}$
D. $\frac{x + 7}{x + 4}$
5. Factorize $216a^3 - 125$.
A. $(6a - 5)(36a^2 - 30a + 25)$
B. $(6a + 5)(36a^2 - 30a + 25)$
C. $(6a - 5)(36a^2 + 30a + 25)$
D. $(6a - 5)(6a^2 + 30a + 25)$
6. If $y = \frac{3x + 7}{x - 4}$, express x in terms of y .
A. $x = \frac{4y - 7}{y - 3}$
B. $x = \frac{4y + 7}{y + 3}$
C. $x = \frac{7 - 4y}{y - 3}$
D. $x = \frac{4y + 7}{y - 3}$
7. Solve $3^x = 80$, correct to 3 significant figures.
A. 3.99
B. 3.98
C. 1.90
D. 26.7
8. Let α and β be the roots of $3x^2 - 11x + 2 = 0$. Find $\alpha^2 + \beta^2$.
A. $\frac{121}{9}$
B. $\frac{109}{9}$
C. $\frac{115}{9}$
D. $\frac{97}{9}$

9. If the equation $x^2 - kx + (5k - 16) = 0$ has two equal real roots, find k .
A. $k = -4$ or $k = -16$
B. $k = 16$ only
C. $k = 8$ or $k = 2$
D. $k = 4$ or $k = 16$
10. Solve $x^2 - 3x - 28 \leq 0$.
A. $x \leq -4$ or $x \geq 7$
B. $-4 \leq x \leq 7$
C. $-7 \leq x \leq 4$
D. $4 \leq x \leq 7$
11. The 2nd and 6th terms of an arithmetic sequence are 11 and 39 respectively. Find the sum of the first 16 terms.
A. 880
B. 904
C. 928
D. 452
12. The geometric sequence is 6, 12, 24, Find the 8th term.
A. 384
B. 768
C. 1536
D. 512
13. Find the remainder when $4x^3 - 2x^2 + 5x - 3$ is divided by $x + 1$.
A. 4
B. 14
C. -6
D. -14
14. It is given that y varies inversely as x^2 . When $x = 2$, $y = 25$. Find the value of y when $x = 5$.
A. 4
B. 10
C. 62.5
D. 100
15. An item is given successive discounts of 20% and 25%. Find the overall percentage discount.
A. 45%
B. 60%
C. 40%
D. 5%
16. The sequence is 10, 17, 24, 31, Find the 16th term.
A. 122
B. 108
C. 115
D. 112
17. \$25000 is deposited at 8% per annum compounded yearly for 2 years. Find the amount, correct to the nearest dollar.
A. \$29000
B. \$27000
C. \$28800
D. \$29160
18. Simplify $\sqrt{128} - \sqrt{32}$.
A. $4\sqrt{2}$
B. $12\sqrt{2}$
C. $\sqrt{96}$
D. $96\sqrt{2}$

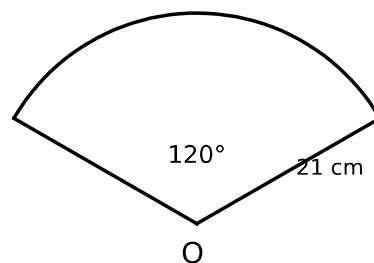
19. In the figure, in $\triangle ABC$, $\angle A = 47^\circ$ and $\angle B = 58^\circ$. Find $\angle ACD$, where D lies on BC produced.



- A. 75°
 B. 105°
 C. 122°
 D. 90°
20. In the figure, $AB \parallel CD$ and a transversal cuts them. The marked angle is 115° . Find x .



- A. 115°
 B. 57.5°
 C. 65°
 D. 25°
21. In the figure, a sector has radius 21 cm and angle at the centre 120° . Find the length of the arc in terms of π .



- A. 7π cm
 B. 14π cm
 C. 147π cm
 D. 42π cm
22. Find the distance between the points (2, 6) and (10, 21).
- A. 17
 B. 23
 C. $\sqrt{23}$
 D. 289

23. Find the slope of a line perpendicular to the line $8x - 3y = 5$.

A. $\frac{8}{3}$
C. $\frac{3}{8}$

B. $-\frac{3}{8}$
D. $-\frac{8}{3}$

24. Find the centre and radius of the circle $x^2 + y^2 + 8x - 14y - 16 = 0$.

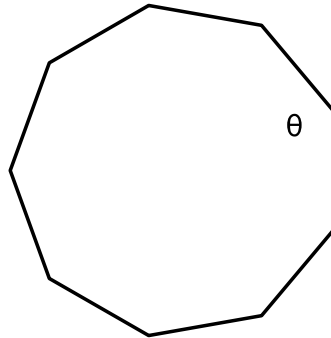
A. centre $(4, -7)$, $r = 9$

B. centre $(-4, 7)$, $r = 81$

C. centre $(-8, 14)$, $r = 9$

D. centre $(-4, 7)$, $r = 9$

25. The figure shows a regular 9-sided polygon. Find each interior angle θ .



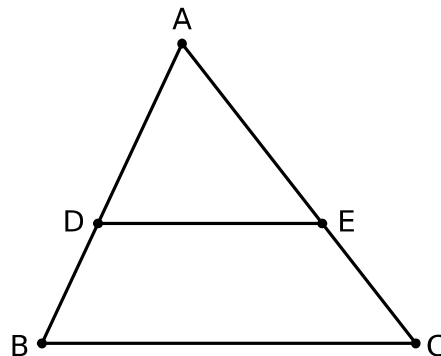
A. 40°

B. 144°

C. 135°

D. 140°

26. In the figure, $DE \parallel BC$. It is given that $AD = 9$, $DB = 6$ and $AE = 15$. Find EC .



A. 10

B. 22.5

C. 6

D. 15

27. Find the volume of a sphere of radius 6 cm in terms of π .

A. $72\pi \text{ cm}^3$

B. $288\pi \text{ cm}^3$

C. $216\pi \text{ cm}^3$

D. $36\pi \text{ cm}^3$

28. If the mean of 11, 17, x , 24 is 20, find x .

A. 20

B. 24

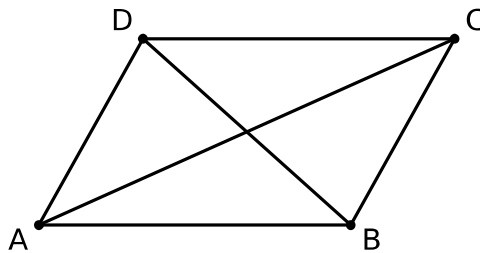
C. 28

D. 26

29. Find the median of the data 5, 8, 11, 13, 17, 20 .
- A. 12
B. 11
C. 13
D. 12.5
30. A fair die is thrown. Find the probability of getting an even number .
- A. $\frac{1}{3}$
B. $\frac{2}{3}$
C. $\frac{1}{2}$
D. $\frac{1}{6}$

Section B

31. Let $a > b$, where a and b are real numbers. I. $a^2 > b^2$ II. $a - b > 0$ III. $-a < -b$ Which of the above is/are true?
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
32. Let x and y be positive numbers. I. $\log \frac{x}{y} = \log x - \log y$ II. $\log(x + y) = \log x \times \log y$ III. $\log(x^4) = 4\log x$ Which of the above is/are true?
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
33. Consider the quadratic function $y = x^2 - 14x + 58$. I. The vertex of the graph is (7, 9) II. The graph opens upwards III. The graph cuts the x -axis at two points Which of the above is/are true?
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
34. Let ABCD be a parallelogram. I. Both pairs of opposite angles are equal II. The diagonals are perpendicular to each other III. $AB \parallel DC$ and $AD \parallel BC$ Which of the above is/are true?

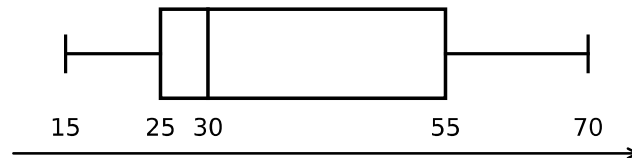


- A. I and II only
B. I and III only
C. II and III only
D. I, II and III

42. In a test, the mean is 62 marks and the standard deviation is 14 marks. A student has a standard score of -1.5 . Find the student's marks.

- A. 41
B. 83
C. 60.5
D. 48

43. The box-and-whisker diagram in the figure shows the data of a distribution. Which of the following about the distribution is correct?



- A. The distribution is negatively skewed
B. The distribution is positively skewed
C. The distribution is symmetric
D. The inter-quartile range is 40

44. A bag contains 6 red balls and 3 white balls. 2 balls are drawn at random at the same time. Find the probability of drawing at least 1 white ball.

- A. $\frac{5}{12}$
B. $\frac{7}{12}$
C. $\frac{1}{12}$
D. $\frac{1}{3}$

45. Find the variance of the data 3, 6, 9, 12, 15.

- A. 18
B. 9
C. $3\sqrt{2}$
D. 90

END OF PAPER